Chapter 10

Modules and Finitely Generated Abelian Groups

We now take a detour from the purely linear-algebraic material to study modules—a generalization of vector spaces where the scalars come from a ring rather than a field. The payoff is immediate: the classification of finitely generated abelian groups reduces to "linear algebra over $\mathbb{Z}$."

Recall from **Lecture 2** that every abelian group $(G, +)$ is a $\mathbb{Z}$-module: the action $\mathbb{Z} \times G \to G$ sends $(n, g) \mapsto ng$ (where $ng = g + g + \cdots + g$ for $n > 0$, $0 \cdot g = 0$, and $ng = -(|n|g)$ for $n < 0$). This observation is the bridge between group theory and module theory.

10.1 Modules over a Ring

10.1.1 Definitions and Examples

Definition 10.1.1: Module

Let R be a ring (with unity 1). A (left) **R-module** is an abelian group $(M, +)$ equipped with a scalar multiplication $R \times M \to M$, $(r, m) \mapsto r \cdot m$, satisfying:

(i) $r(m_1 + m_2) = rm_1 + rm_2$ for all $r \in R$, $m_1, m_2 \in M$.

(ii) $(r_1 + r_2)m = r_1 m + r_2 m$ for all $r_1, r_2 \in R$, $m \in M$.

(iii) $(r_1 r_2)m = r_1(r_2 m)$ for all $r_1, r_2 \in R$, $m \in M$.

(iv) $1 \cdot m = m$ for all $m \in M$.

Example 10.1.1 (Vector spaces)

If $R = k$ is a field, then a k-module is the same thing as a vector space over k. So modules generalize vector spaces by allowing the scalars to come from a ring.

Example 10.1.2 (Abelian groups as $\mathbb{Z}$-modules)

Every abelian group $(G, +)$ is a $\mathbb{Z}$-module via $n \cdot g = ng$. Conversely, every $\mathbb{Z}$-module is an abelian group (just forget the $\mathbb{Z}$-action, which is determined by the group structure anyway). The categories of abelian groups and $\mathbb{Z}$-modules are the same.

Example 10.1.3 (The ring as a module over itself)

R is an R-module via left multiplication: $r \cdot s = rs$. This is a free module of rank 1.

Example 10.1.4 (R-modules from ring homomorphisms)

If $\varphi : R \to S$ is a ring homomorphism, then any S-module M becomes an R-module via $r \cdot m = \varphi(r)m$. This is called restriction of scalars.

Example 10.1.5 (Polynomial modules)

Let $R = k[x]$ and V a vector space over k. Any linear operator $T : V \to V$ makes V into a $k[x]$-module via $p(x) \cdot v = p(T)(v)$. Different operators give different module structures on the same underlying vector space.

10.1.2 Submodules and Ideals

Definition 10.1.2: Submodule

A **submodule** of an R-module M is a subset $N \subseteq M$ that is itself an R-module under the inherited operations. Equivalently, N is a subgroup of $(M, +)$ that is closed under scalar multiplication: $rn \in N$ for all $r \in R$, $n \in N$.

Definition 10.1.3: Ideal

A submodule of R (viewed as a module over itself) is called an **ideal**. That is, an ideal $I \subseteq R$ is a subset satisfying:

(i) I is an abelian subgroup of $(R, +)$.

(ii) $R \cdot I \subseteq I$: for all $r \in R$ and $a \in I$, we have $ra \in I$.

Example 10.1.6 (Ideals in $\mathbb{Z}$ and $k[x]$)

The ideals of $\mathbb{Z}$ are exactly the subgroups $n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$ for $n \geqslant 0$. Similarly, the ideals of $k[x]$ are $p(x) \cdot k[x] = \{p(x)q(x) : q(x) \in k[x]\}$. In both cases, every ideal is generated by a single element. Rings with this property are called **principal ideal domains** (PIDs). This is closely related to the Euclidean division algorithm: $\text{span}(p, q) = \text{span}(\gcd(p, q))$.

Note:-

The quotient of an R-module M by a submodule N is again an R-module, with $r \cdot (m + N) = rm + N$. For example, $\mathbb{Z}/n\mathbb{Z} = \mathbb{Z}/n$ as a $\mathbb{Z}$-module, and $k[x]/(p(x)) \cdot k[x]$ is a $k[x]$-module. The quotient of R by an ideal is not just an R-module but a ring in its own right.

10.1.3 General Facts about Modules: Nothing Is True

Over a field, every vector space has a basis, every linearly independent set extends to one, and every spanning set contains one. Over a general ring, all of these fail.

Example 10.1.7 (Bases need not exist)

The $\mathbb{Z}$-module $M = \mathbb{Z}/n$ (for $n \geqslant 2$) has no basis. Any map $\varphi : \mathbb{Z} \to M$ sending $1 \mapsto m$ cannot be injective (since $nm = 0$ in M but $n \neq 0$ in $\mathbb{Z}$). So no single-element set is linearly independent, and no basis can exist.

Example 10.1.8 (Linearly independent sets may not be part of a basis)

Consider $M = \mathbb{Z}$ as a $\mathbb{Z}$-module. The element $2 \in \mathbb{Z}$ is linearly independent (if $a \cdot 2 = 0$ then $a = 0$). But $\{2\}$ cannot be extended to a basis of $\mathbb{Z}$: the only bases of $\mathbb{Z}$ are $\{1\}$ and $\{-1\}$, and no set containing 2 is a basis.

Example 10.1.9 (Spanning sets need not contain a basis)

The $\mathbb{Z}$-module $M = \mathbb{Z}$ is spanned by $\{4, 5\}$ (since $\gcd(4, 5) = 1$, so $1 = 5 \cdot 5 - 6 \cdot 4 \in \mathrm{span}(4, 5)$). But $\{4, 5\}$ is not linearly independent ($5 \cdot 4 - 4 \cdot 5 = 0$), and neither subset $\{4\}$ or $\{5\}$ spans $\mathbb{Z}$. So the spanning set contains no basis.

Example 10.1.10 (Submodules of finitely generated modules need not be finitely generated)

Let $R = k[x_1, x_2, \ldots]$ (polynomials in infinitely many variables) and $M = R$ as an R-module. Then M is generated by $\{1\}$. But the ideal $M' = \{f \in R : f(0) = 0\}$ (polynomials with zero constant term) is a submodule not finitely generated: any finite subset only involves finitely many variables x_i, so cannot span x_j for large j.
By contrast, this pathology does not occur over Noetherian rings (including $\mathbb{Z}$ and $k[x_1, \ldots, x_n]$).

10.1.4 Module Homomorphisms

Definition 10.1.4: Module homomorphism

Let M, N be R-modules. A map $\varphi \in \mathrm{Hom}_R(M, N)$ is a module homomorphism if

$$\varphi(u + w) = \varphi(u) + \varphi(w) \quad \text{and} \quad \varphi(av) = a\varphi(v)$$

for all $u, w \in M$, $a \in R$.

Note:-

$\mathrm{Hom}_R(M, N)$ is itself an R-module: $(\varphi + \psi)(v) = \varphi(v) + \psi(v)$ and $(a\varphi)(v) = a\varphi(v)$. For free modules, things work as expected: $\mathrm{Hom}_R(R^m, R^n) \cong R^{m \times n}$ (determined by images of basis vectors, just like linear maps between vector spaces). But for non-free modules, $\mathrm{Hom}_R(M, N)$ can vanish even when both M and N are nonzero. For instance, $\mathrm{Hom}_R(k, k[x]) = 0$ when $R = k[x]$ and $M = k = k[x]/(x)$: any φ must satisfy $x \cdot \varphi(1) = \varphi(x \cdot 1) = \varphi(0) = 0$, so $\varphi(1)$ is killed by x in $k[x]$, forcing $\varphi(1) = 0$.

10.2 Classification of Finitely Generated Abelian Groups

The main theorem of this lecture is:

Theorem 10.2.1 Classification of finitely generated abelian groups

Every finitely generated abelian group G is isomorphic to a product of cyclic groups:

$$G \cong \mathbb{Z}/n_1 \times \mathbb{Z}/n_2 \times \cdots \times \mathbb{Z}/n_k \times \mathbb{Z}^\ell$$

where $n_i \geqslant 2$ and $\ell \geqslant 0$. Moreover, using the Chinese Remainder Theorem ($\mathbb{Z}/mn \cong \mathbb{Z}/m \times \mathbb{Z}/n$ when $\gcd(m, n) = 1$), we can arrange each n_i to be a prime power. In this form, the decomposition is unique up to reordering.

The proof reduces to "linear algebra over $\mathbb{Z}$," following Artin §14.4–14.7. The strategy has two ingredients: first, express any finitely generated $\mathbb{Z}$-module as $\mathbb{Z}^n/\mathrm{Im}(T)$ for some $T \in \mathrm{Hom}(\mathbb{Z}^m, \mathbb{Z}^n)$; second, diagonalize T by changes of basis over $\mathbb{Z}$.

10.2.1 Step 1: Presenting a Module as a Quotient

Proposition 10.2.1 Presentation of finitely generated $\mathbb{Z}$-modules

If M is a finitely generated $\mathbb{Z}$-module, then there exist $m, n \geqslant 0$ and $T \in \mathrm{Hom}(\mathbb{Z}^m, \mathbb{Z}^n)$ such that

$$M \cong \mathbb{Z}^n / \mathrm{Im}(T).$$

Equivalently, there is an exact sequence $\mathbb{Z}^m \xrightarrow{T} \mathbb{Z}^n \to M \to 0$.

This says: quotient by a $\mathbb{Z}$-submodule of $\mathbb{Z}^n$ is the same as quotient of an abelian group by a subgroup. We need a lemma:

Lemma 10.2.1 Submodules of $\mathbb{Z}^n$ are free

Any submodule of $\mathbb{Z}^n$ is finitely generated and free of rank $\leqslant n$.

Proof: By induction on n. For $n = 1$: subgroups of $(\mathbb{Z}, +)$ are $\{0\}$ or $\mathbb{Z}a$ for some $a \in \mathbb{Z} \setminus \{0\}$, both free of rank $\leqslant 1$.

Assume the result for $\mathbb{Z}^{n-1}$, and let $N \subseteq \mathbb{Z}^n$ be a submodule. The projection $\pi : \mathbb{Z}^n \to \mathbb{Z}^{n-1}$ sending $(a_1, \dots, a_n) \mapsto (a_2, \dots, a_n)$ restricts to a homomorphism $\pi|_N : N \to \mathbb{Z}^{n-1}$.

The image $\mathrm{Im}(\pi|_N)$ is a submodule of $\mathbb{Z}^{n-1}$, hence free and finitely generated by induction. The kernel $\mathrm{Ker}(\pi|_N) = N \cap (\mathbb{Z} \times 0 \times \cdots \times 0)$ is a subgroup of $\mathbb{Z}$, hence free of rank 0 or 1.

If $\mathrm{Ker}(\pi|_N)$ and $\mathrm{Im}(\pi|_N)$ are both finitely generated (and free), then so is N. Let $e_1, \dots, e_k$ be generators of $\mathrm{Ker}(\pi|_N)$ (a basis) and $g_1 = \pi(f_1), \dots, g_m = \pi(f_m)$ generators of $\mathrm{Im}(\pi|_N)$. For any $x \in N$, write $\pi(x) = \sum a_i g_i$, so $\pi(x - \sum a_i f_i) = 0$, meaning $x - \sum a_i f_i \in \mathrm{Ker}(\pi|_N) = \mathrm{span}(e_1, \dots, e_k)$. Thus $x \in \mathrm{span}(e_1, \dots, e_k, f_1, \dots, f_m)$, so $\{e_i, f_j\}$ generates N. That this set is a basis (i.e., free) follows from the directness of $\mathrm{Ker} \cap \mathrm{Im} = 0$. $\square$

Proof of the proposition: Let M be finitely generated with generators $e_1, \dots, e_n$. Define $\varphi : \mathbb{Z}^n \to M$ by $(a_1, \dots, a_n) \mapsto \sum a_i e_i$. This is surjective. The kernel $N = \mathrm{Ker}(\varphi) \subseteq \mathbb{Z}^n$ is a submodule, hence finitely generated by the lemma. Let $f_1, \dots, f_m$ be generators of N. Define $T : \mathbb{Z}^m \to \mathbb{Z}^n$ by $T(b_i) = f_i$. Then $\mathrm{Ker}(\varphi) = \mathrm{Im}(T)$, giving the exact sequence

$$\mathbb{Z}^m \xrightarrow{T} \mathbb{Z}^n \xrightarrow{\varphi} M \to 0$$

and thus $M \cong \mathbb{Z}^n / \mathrm{Im}(T)$. $\square$

10.2.2 Step 2: Diagonalization over $\mathbb{Z}$ (Smith Normal Form)

The key difference from linear algebra over a field: we cannot divide freely in $\mathbb{Z}$. The substitute is the Euclidean algorithm, which governs what "row and column operations" are available.

Definition 10.2.1: Divisibility

The **divisibility** of a nonzero element $x = (a_1, \dots, a_n) \in \mathbb{Z}^n$ is the largest $d \in \mathbb{Z}_+$ such that $x = dy$ for some $y \in \mathbb{Z}^n$. Equivalently, $d = \gcd(a_1, \dots, a_n)$. An element is **primitive** if its divisibility is 1.

Lemma 10.2.2 Primitive elements and bases

An element of $\mathbb{Z}^n$ can be part of a basis if and only if it is primitive. More generally, $v \in \mathbb{Z}^n$ with divisibility d satisfies $v = d \cdot w$ where w is a basis element.

Proof: Elements of a basis $(e_1, \dots, e_n)$ are primitive: linear independence prevents $e_i = d(\sum a_j e_j)$ for $d > 1$.

Conversely, let $v = a_1e_1 + \cdots + a_ne_n$ be primitive. Without loss of generality, $a_1 \neq 0$ and $|a_1| = \min\{|a_i| : a_i \neq 0\}$. For each $k \geqslant 2$, write $a_k = q_ka_1 + r_k$ by Euclidean division. Change basis to $(e_1' = e_1 + \sum_{k\geqslant 2} q_ke_k,\ e_2, \ldots, e_n)$. In this basis, $v = a_1e_1' + r_2e_2 + \cdots + r_ne_n$ with $0 \leqslant r_k < |a_1|$.

Repeating this process (which strictly decreases the minimum nonzero coefficient at each step) terminates in finitely many steps with $v = d \cdot e_1''$ for some basis vector e_1''. Since v is primitive, $d = 1$. □

Theorem 10.2.2 Smith Normal Form

For any $T \in \mathrm{Hom}(\mathbb{Z}^m, \mathbb{Z}^n)$, there exist bases $(e_1, \ldots, e_m)$ of $\mathbb{Z}^m$ and $(f_1, \ldots, f_n)$ of $\mathbb{Z}^n$, and positive integers $d_1, \ldots, d_r$ (where $r \leqslant \min(m, n)$ is the rank of T), such that

$$T(e_i) = \begin{cases} d_if_i & \text{if } 1 \leqslant i \leqslant r \\ 0 & \text{if } i > r. \end{cases}$$

In matrix form:

$$\mathcal{M}(T) = \begin{pmatrix} d_1 & & & 0 & \\ & d_2 & & & \\ & & \ddots & & \\ & & & d_r & \\ 0 & & & & 0 \end{pmatrix}.$$

Moreover, the construction gives $d_1 = \min\{\mathrm{div}(T(x)) : x \notin \mathrm{Ker}(T)\} = \gcd$ of all entries of $\mathcal{M}(T)$.

Proof: By induction on m.

If $T = 0$, the statement is trivial for any m, n.

Case $m = 1$: T is determined by $T(1) \in \mathbb{Z}^n$. Let $d = \mathrm{div}(T(1))$. By the lemma, $T(1) = df_1$ where f_1 can be extended to a basis of $\mathbb{Z}^n$. Take $e_1 = 1$.

Inductive step: Assume $T \neq 0$. Let $d_1 = \min\{\mathrm{div}(T(x)) : x \notin \mathrm{Ker}(T)\}$, and choose e_1 with $\mathrm{div}(T(e_1)) = d_1$. Since $T(e_1)$ has divisibility d_1, write $T(e_1) = d_1f_1$ with f_1 primitive, and extend f_1 to a basis $(f_1, \ldots, f_n)$ of $\mathbb{Z}^n$. Extend e_1 to a basis $(e_1, \ldots, e_m)$ of $\mathbb{Z}^m$ (using the lemma, since e_1 is primitive: if $e_1 = d \cdot e_1'$ then $\mathrm{div}(T(e_1')) = d_1/d$, contradicting minimality unless $d = 1$).

In the bases (e_i) and (f_j), the matrix of T has the form

$$\mathcal{M}(T) = \begin{pmatrix} d_1 & a_2 & \cdots & a_m \\ 0 & & & \\ \vdots & & \mathcal{M}(T') & \\ 0 & & & \end{pmatrix}$$

where T' is the restriction to $\mathrm{span}(e_2, \ldots, e_m)$ composed with projection to $\mathrm{span}(f_2, \ldots, f_n)$.

For $j \geqslant 2$, write $a_j = q_jd_1 + r_j$ with $0 \leqslant r_j < d_1$. Change basis of $\mathbb{Z}^m$ to $(e_1, e_2' = e_2 - q_2e_1, \ldots, e_m' = e_m - q_me_1)$. The matrix becomes

$$\begin{pmatrix} d_1 & r_2 & \cdots & r_m \\ 0 & & & \\ \vdots & & \mathcal{M}(T') & \\ 0 & & & \end{pmatrix}.$$

If any $r_j \neq 0$, then $\mathrm{div}(T(e_j')) \leqslant r_j < d_1$, contradicting the minimality of d_1. So $r_j = 0$ for all $j \geqslant 2$, and the matrix is

$$\begin{pmatrix} d_1 & 0 & \cdots & 0 \\ 0 & & & \\ \vdots & & \mathcal{M}(T') & \\ 0 & & & \end{pmatrix}.$$

By induction on m, we can diagonalize T' by further basis changes in $\mathrm{span}(e_2, \ldots, e_m)$ and $\mathrm{span}(f_2, \ldots, f_n)$, giving the desired diagonal form. □

Note:-

The Smith Normal Form is the integer analogue of row reduction. Over a field, Gaussian elimination diagonalizes any matrix (in fact reduces it to have 1's and 0's on the diagonal). Over $\mathbb{Z}$, the Euclidean algorithm plays the role of division, and the diagonal entries d_i need not be 1. The construction actually forces $d_1 \mid d_2 \mid \cdots \mid d_r$ (the invariant factor divisibility chain), though we do not prove this here.

10.2.3 Step 3: Completing the Classification

Proof of the classification theorem: By Proposition 1 (presentation as quotient), any finitely generated abelian group M satisfies $M \cong \mathbb{Z}^n / \operatorname{Im}(T)$ for some $T \in \operatorname{Hom}(\mathbb{Z}^m, \mathbb{Z}^n)$. By the Smith Normal Form, after choosing appropriate bases, $\operatorname{Im}(T)$ is spanned by $d_1 f_1, \ldots, d_r f_r$ for some positive integers $d_i > 0$ and $r \leqslant n$. Therefore

$$M \cong \mathbb{Z}^n / \operatorname{span}(d_1 f_1, \ldots, d_r f_r) \cong \mathbb{Z}/d_1 \times \cdots \times \mathbb{Z}/d_r \times \mathbb{Z}^{n-r}.$$

The factors $\mathbb{Z}/d_i$ with $d_i = 1$ are trivial and can be dropped. Relabeling the remaining $d_i \geqslant 2$ as $n_1, \ldots, n_k$ and setting $\ell = n - r$ gives the decomposition

$$M \cong \mathbb{Z}/n_1 \times \cdots \times \mathbb{Z}/n_k \times \mathbb{Z}^\ell.$$

By the Chinese Remainder Theorem, each $\mathbb{Z}/n_i$ with $n_i = p_1^{a_1} \cdots p_s^{a_s}$ decomposes as $\mathbb{Z}/p_1^{a_1} \times \cdots \times \mathbb{Z}/p_s^{a_s}$. Rearranging gives the prime power form. □

Example 10.2.1 (Finitely generated abelian groups of small order)

- Order 4: $\mathbb{Z}/4$ or $\mathbb{Z}/2 \times \mathbb{Z}/2$. These are not isomorphic (the first has an element of order 4, the second does not).
- Order 6: $\mathbb{Z}/6 \cong \mathbb{Z}/2 \times \mathbb{Z}/3$ (only one, since $\gcd(2, 3) = 1$).
- Order 8: $\mathbb{Z}/8$, $\mathbb{Z}/4 \times \mathbb{Z}/2$, or $\mathbb{Z}/2 \times \mathbb{Z}/2 \times \mathbb{Z}/2$.
- Order 12: $\mathbb{Z}/12 \cong \mathbb{Z}/4 \times \mathbb{Z}/3$, or $\mathbb{Z}/2 \times \mathbb{Z}/6 \cong \mathbb{Z}/2 \times \mathbb{Z}/2 \times \mathbb{Z}/3$.

Note:-

The number ℓ is the **free rank** (or **Betti number**) of M, and the integers $n_1, \ldots, n_k$ (in prime power form) are the **elementary divisors**. These invariants completely determine M up to isomorphism. The free rank counts the number of "copies of $\mathbb{Z}$," while the elementary divisors encode the torsion (elements of finite order).

Exercise 10.2.1 Modules and finitely generated abelian groups exercises

(1) Classify all abelian groups of order 36 up to isomorphism.

(2) Let $T : \mathbb{Z}^2 \to \mathbb{Z}^2$ be the linear map with matrix $\begin{pmatrix} 6 & 4 \\ 3 & 5 \end{pmatrix}$. Compute the Smith Normal Form of T and determine the group $\mathbb{Z}^2 / \operatorname{Im}(T)$.

(3) Prove that a finitely generated abelian group is free (i.e., isomorphic to $\mathbb{Z}^n$ for some n) if and only if it is torsion-free (i.e., the only element of finite order is 0).

(4) Let G be a finitely generated abelian group with $|G| = p^n$ for a prime p. Show that $G \cong \mathbb{Z}/p^{a_1} \times \cdots \times \mathbb{Z}/p^{a_k}$ where $a_1 + \cdots + a_k = n$ and $a_i \geqslant 1$. How many such groups are there (as a function of the partition structure of n)?

(5) Show that $\mathbb{Z}/m \otimes_{\mathbb{Z}\mathbb{Z}/n \cong \mathbb{Z}/\gcd(m,n)}$. (Hint: use the presentation $\mathbb{Z}/m = \mathbb{Z}/m\mathbb{Z}$ and the right-exactness of the tensor product.)

Solution

(1) We have $36 = 2^2 \cdot 3^2$. For the 2-part: $\mathbb{Z}/4$ or $\mathbb{Z}/2 \times \mathbb{Z}/2$. For the 3-part: $\mathbb{Z}/9$ or $\mathbb{Z}/3 \times \mathbb{Z}/3$. Taking all combinations:

$$\mathbb{Z}/4 \times \mathbb{Z}/9 \cong \mathbb{Z}/36, \quad \mathbb{Z}/4 \times \mathbb{Z}/3 \times \mathbb{Z}/3, \quad \mathbb{Z}/2 \times \mathbb{Z}/2 \times \mathbb{Z}/9, \quad \mathbb{Z}/2 \times \mathbb{Z}/2 \times \mathbb{Z}/3 \times \mathbb{Z}/3.$$

So there are exactly 4 abelian groups of order 36.

(2) Start with $A = \begin{pmatrix} 6 & 4 \\ 3 & 5 \end{pmatrix}$. The gcd of all entries is 1, but let us apply row/column operations over $\mathbb{Z}$.

$R_1 \to R_1 - 2R_2$: $\begin{pmatrix} 0 & -6 \\ 3 & 5 \end{pmatrix}$. Swap rows: $\begin{pmatrix} 3 & 5 \\ 0 & -6 \end{pmatrix}$. $C_2 \to C_2 - C_1$: $\begin{pmatrix} 3 & 2 \\ 0 & -6 \end{pmatrix}$. $C_1 \to C_1 - C_2$: $\begin{pmatrix} 1 & 2 \\ 6 & -6 \end{pmatrix}$. $C_2 \to C_2 - 2C_1$: $\begin{pmatrix} 1 & 0 \\ 6 & -18 \end{pmatrix}$. $R_2 \to R_2 - 6R_1$: $\begin{pmatrix} 1 & 0 \\ 0 & -18 \end{pmatrix}$.

Smith Normal Form: $\operatorname{diag}(1, 18)$. So $\mathbb{Z}^2/\operatorname{Im}(T) \cong \mathbb{Z}/1 \times \mathbb{Z}/18 \cong \mathbb{Z}/18 \cong \mathbb{Z}/2 \times \mathbb{Z}/9$.

Check: $\det(A) = 30 - 12 = 18$, and $|\mathbb{Z}^2/\operatorname{Im}(T)| = |\det(T)| = 18$. Consistent.

(3) One direction: if $G \cong \mathbb{Z}^n$, then G is torsion-free (only $0 \in \mathbb{Z}^n$ has finite order).

Conversely, if G is finitely generated and torsion-free, write $G \cong \mathbb{Z}/n_1 \times \cdots \times \mathbb{Z}/n_k \times \mathbb{Z}^\ell$ by the classification theorem. Each $\mathbb{Z}/n_i$ contributes a nonzero element of finite order (namely $\bar{1}$, which has order n_i). Since G is torsion-free, $k = 0$, so $G \cong \mathbb{Z}^\ell$.

(4) Since $|G| = p^n$, the classification gives $G \cong \mathbb{Z}/p^{a_1} \times \cdots \times \mathbb{Z}/p^{a_k}$ with $a_i \geqslant 1$ and $p^{a_1} \cdots p^{a_k} = p^n$, i.e., $a_1 + \cdots + a_k = n$. The number of such groups equals the number of partitions of n. For example, p^4 gives partitions $(4), (3,1), (2,2), (2,1,1), (1,1,1,1)$, so 5 groups.

(5) From the short exact sequence $0 \to \mathbb{Z} \xrightarrow{\times m} \mathbb{Z} \to \mathbb{Z}/m \to 0$, tensor with $\mathbb{Z}/n$. Right-exactness gives:

$$\mathbb{Z}/n \xrightarrow{\times m} \mathbb{Z}/n \to \mathbb{Z}/m \otimes \mathbb{Z}/n \to 0.$$

The map $\times m : \mathbb{Z}/n \to \mathbb{Z}/n$ has image $m\mathbb{Z}/n \cong \mathbb{Z}/\gcd(m,n)$ (the image of multiplication by m in $\mathbb{Z}/n$ is the cyclic subgroup of order $n/\gcd(m,n)$). So:

$$\mathbb{Z}/m \otimes \mathbb{Z}/n \cong (\mathbb{Z}/n)/(m\mathbb{Z}/n) \cong \mathbb{Z}/\gcd(m,n).$$